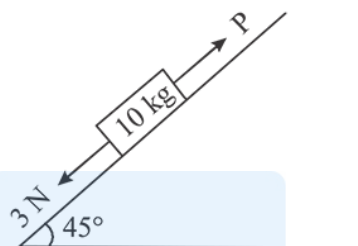


JEE Mains 2019 Chapter wise Question Bank

Newton's Laws of Motion - Questions

Q1

A block of mass 10 kg is kept on a rough inclined plane as shown in the figure. A force of 3 N is applied on the block. The coefficient of static friction between the plane and the block is 0.6. What should be the minimum value of force P, such that the block does not move downward? (take $g = 10 \text{ ms}^{-2}$)



- (1) 32 N (2) 18 N
(3) 23 N (4) 25 N

9 Jan Morning

Q2

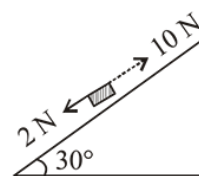
A particle of mass m is moving in a straight line with momentum p . Starting at time $t = 0$, a force $F = kt$ acts in the same direction on the moving particle during time interval T so that its momentum changes from p to $3p$. Here k is a constant. The value of T is :

- (1) $2\sqrt{\frac{k}{p}}$ (2) $2\sqrt{\frac{p}{k}}$
(3) $\sqrt{\frac{2k}{p}}$ (4) $\sqrt{\frac{2p}{k}}$

11 Jan Evening

Q3

A block kept on a rough inclined plane, as shown in the figure, remains at rest upto a maximum force 2 N down the inclined plane. The maximum external force up the inclined plane that does not move the block is 10 N. The coefficient of static friction between the block and the plane is : [Take $g = 10 \text{ m/s}^2$]



- (1) $\frac{\sqrt{3}}{2}$ (2) $\frac{\sqrt{3}}{4}$
(3) $\frac{1}{2}$ (4) $\frac{2}{3}$

12 Jan Evening

Q4

A ball is thrown upward with an initial velocity V_0 from the surface of the earth. The motion of the ball is affected by a drag force equal to $m\gamma v^2$ (where m is mass of the ball, v is its instantaneous velocity and γ is a constant). Time taken by the ball to rise to its zenith is :

- (1) $\frac{1}{\sqrt{\gamma g}} \tan^{-1} \left(\sqrt{\frac{\gamma}{g}} V_0 \right)$ (2) $\frac{1}{\sqrt{\gamma g}} \sin^{-1} \left(\sqrt{\frac{\gamma}{g}} V_0 \right)$
(3) $\frac{1}{\sqrt{\gamma g}} \ln \left(1 + \sqrt{\frac{\gamma}{g}} V_0 \right)$ (4) $\frac{1}{\sqrt{2\gamma g}} \tan^{-1} \left(\sqrt{\frac{2\gamma}{g}} V_0 \right)$

10 April Morning

Q5

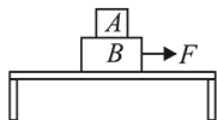
A bullet of mass 20g has an initial speed of 1 ms^{-1} , just before it starts penetrating a mud wall of thickness 20 cm. If the wall offers a mean resistance of $2.5 \times 10^{-2} \text{ N}$, the speed of the bullet after emerging from the other side of the wall is close to :

- (1) 0.1 ms^{-1} (2) 0.7 ms^{-1} (3) 0.3 ms^{-1} (4) 0.4 ms^{-1}

10 April Evening

Q6

Two blocks A and B masses $m_A = 1 \text{ kg}$ and $m_B = 3 \text{ kg}$ are kept on the table as shown in figure. The coefficient of friction between A and B is 0.2 and between B and the surface of the table is also 0.2. The maximum force F that can be applied on B horizontally, so that the block A does not slide over the block B is : [Take $g = 10 \text{ m/s}^2$]

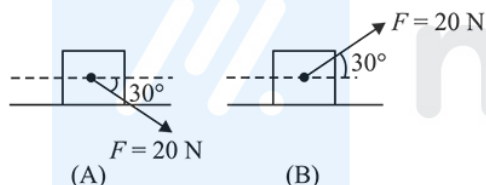


- (1) 8 N (2) 16 N (3) 40 N (4) 12 N

10 April Evening

Q7

A block of mass 5 kg is (i) pushed in case (A) and (ii) pulled in case (B), by a force $F = 20 \text{ N}$, making an angle of 30° with the horizontal, as shown in the figures. The coefficient of friction between the block and floor is $\mu = 0.2$. The difference between the accelerations of the block, in case (B) and case (A) will be : ($g = 10 \text{ ms}^{-2}$)



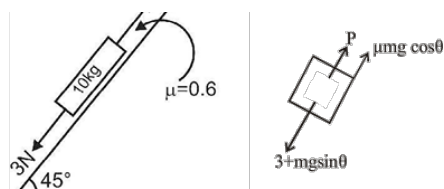
- (1) 0.4 ms^{-2} (2) 3.2 ms^{-2}
 (3) 0.8 ms^{-2} (4) 0 ms^{-2}

12 April Evening

JEE Mains 2019 Chapter wise Question Bank

Newton's Laws of Motion - Answers

Q1



$$mg \sin 45^\circ = \frac{100}{\sqrt{2}} = 50\sqrt{2}$$

$$[\because m=10\text{kg}, g=9.8\text{ms}^{-2}]$$

$$\mu mg \cos \theta = 0.6 \times mg \times \frac{1}{\sqrt{2}} = 0.6 \times 50\sqrt{2}$$

$$3 + mg \sin \theta = P + \mu mg \cos \theta$$

$$3 + 50\sqrt{2} = P + 30\sqrt{2}$$

$$\therefore P = 31.28 = 32 \text{ N}$$

9 Jan Morning

Q2

(2) From Newton's second law

$$\frac{dp}{dt} = F = kt$$

Integrating both sides we get,

$$\int_p^{3p} dp = \int_0^T kt dt \Rightarrow [p]_p^{3p} = k \left[\frac{t^2}{2} \right]_0^T$$

$$\Rightarrow 2p = \frac{kT^2}{2} \Rightarrow T = 2\sqrt{\frac{p}{k}}$$

11 Jan Evening

Q3

(1) From figure, $2 + mg \sin 30^\circ = \mu mg \cos 30^\circ$ and

$$10 = mg \sin 30^\circ + \mu mg \cos 30^\circ$$

$$= 2\mu mg \cos 30^\circ - 2$$

$$\Rightarrow 6 = \mu mg \cos 30^\circ \text{ and}$$

$$4 = mg \cos 30^\circ$$

By dividing above two

$$\Rightarrow \frac{3}{2} = \mu \times \sqrt{3}$$

$$\therefore \text{Coefficient of friction, } \mu = \frac{\sqrt{3}}{2}$$

12 Jan Evening

Q4

(1) Net acceleration

$$\frac{dv}{dt} = a = -(g + \gamma v^2)$$

Let time t required to rise to its zenith ($v = 0$) so,

$$\Rightarrow \int_{v_0}^0 \frac{-dv}{g + \gamma v^2} = \int_0^t dt \quad [\text{for } H_{\text{max}}, v = 0]$$

$$\Rightarrow t = \frac{1}{\sqrt{\gamma g}} \tan^{-1} \left(\frac{\sqrt{\gamma} v_0}{\sqrt{g}} \right)$$

10 April Morning

Q5

Newton's Laws of Motion

(2) From the third equation of motion

$$v^2 - u^2 = 2aS$$

$$\text{But, } a = \frac{F}{m}$$

$$\therefore v^2 = u^2 - 2\left(\frac{F}{m}\right)S$$

$$\Rightarrow v^2 = (1)^2 - (2)\left[\frac{2.5 \times 10^{-2}}{20 \times 10^{-3}}\right] \frac{20}{100}$$

$$\Rightarrow v^2 = 1 - \frac{1}{2}$$

$$\Rightarrow v = \frac{1}{\sqrt{2}} \text{ m/s} = 0.7 \text{ m/s}$$

10 April Evening

Q6

(2) Taking (A + B) as system

$$F - \mu(M + m)g$$

$$= (M + m)a$$

$$\Rightarrow a = \frac{F - \mu(M + m)g}{(M + m)}$$

$$a = \frac{F - (0.2)4 \times 10}{4} = \left(\frac{F - 8}{4}\right) \quad \dots(i)$$

$$\text{But, } a_{\max} = \mu g = 0.2 \times 10 = 2$$

$$\therefore \frac{F - 8}{4} = 2$$

$$\Rightarrow F = 16 \text{ N}$$

10 April Evening

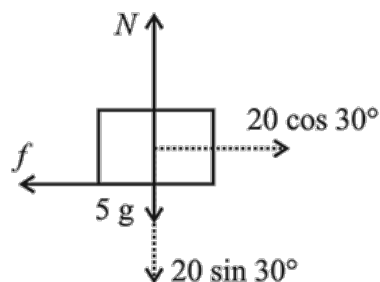
Q7

JEE Mains 2019 Chapter wise Question Bank

(3) A : $N = 5g + 20 \sin 30^\circ$

$$= 50 + 20 \times \frac{1}{2}$$

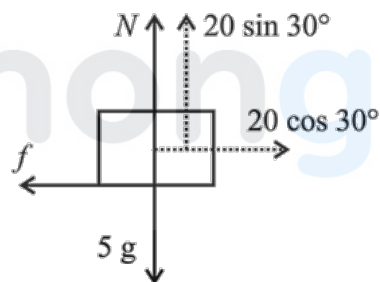
$$= 60 \text{ N}$$



$$\text{Acceleration, } a_1 = \frac{F - f}{m} = \frac{20 \cos 30^\circ - \mu N}{5}$$

$$= \left[\frac{20 \times \frac{\sqrt{3}}{2} - 0.2 \times 60}{5} \right]$$

$$= 1.06 \text{ m/s}^2$$



B : $N = 5g - 20 \sin 30^\circ$

$$= 50 - 20 \times \frac{1}{2} = 40 \text{ N}$$

$$a_2 = \frac{F - f}{m} = \left[\frac{20 \cos 30^\circ - 0.2 \times 40}{5} \right]$$

$$= 1.86 \text{ m/s}^2$$

$$\text{Now } a_2 - a_1 = 1.86 - 1.06$$

$$= 0.8 \text{ m/s}^2$$

12 April Evening